Psychrometric Tool

Example: Enthalpy Calculation · built Nov 12, 2026

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Enthalpy can be calculated from known Dry Bulb Temperature (DBT), Relative Humidity (RH) and Atmospheric Pressure. Steps to determine enthalpy are listed below:

1. Calculate Dew point from DBT, RH and atmospheric pressure

$$T_d = 243.04 \frac{\log\frac{RH}{100} + \frac{(17.625 + DBT)}{(243.04 + DBT)}}{17.625 - \log\frac{RH}{100} - \frac{(17.625 + DBT)}{(243.04 + DBT)}}$$

where

Td= Dew point temperature in degree centigrate

LN = Natural logarithm

RH= Relative HUmidity in percent

DBT= Dry Bulb Temperature

Atmosphere = Atmospheric pressure, [Hpa]

2. Calculate Water Vapor Saturation Pressure (Pws) from the dew point temperature

$$P_{ws} = A.10^{\frac{m.T_d}{(T_d + T_n)}}$$

where ,

Pws = Partial Pressure of water at saturation at given temperature, [hPa]

Td = Dewpoint temperature

A, m, Tn = curve fitting constants which are obtained as below

	A	m	Tn	max error	Temperature r
water	6.116441	7.591386	240.7263	0.083%	-20...+50°C
	6.004918	7.337936	229.3975	0.017%	+50...+100°C
	5.856548	7.27731	225.1033	0.003%	+100...+150°C
	6.002859	7.290361	227.1704	0.007%	+150...+200°(
	9.980622	7.388931	263.1239	0.395%	+200...+350°(

3. Calculate the mixing ratio (X), mass of water to mass of air

$$X = B.\frac{P_{ws}}{(P_{atm} - P_{ws})}$$

where.

X= mixing ration, [g/Kg]

Pws = Partial Pressure of water at saturation at given temperature, [hPa]

Patm = Atmospheric pressure, [hPa]

B = Molecular mass of water over the molecular weight of gas x 1000 , 621.9907 for air [g/kg]

4. Calculate the enthalpy

$$h = DBT.(1.01 + 0.00189X) + 2.5 \quad (kJ/kg)$$

Note: TBD: Below zero degrees, use different formula for dewpoint.

Example

Let's see one example here. Lets assume your outdoor environment has 20 degree centigrate with 80% RH. Consider standard pressure 1013 Hpa. We will calculate all other parameters and enthalpy finally.

So, given:

$$DBT = 20\ °C, RH = 80\%, P_{atm} = 1013HPa$$

Using above mentioned formula, we will have following:

$$T_d = 16.44°C$$

$$P_{ws} = 18.70\ hPa,\ using\ A = 6.116441, m = 7.591386, T_n = 240.7264\ from\ table$$

$$X = 11.70\ g/kg\ , using\ B = 621.9907\ g/kg\ , P_w = P_{ws} = 18.70$$

$$h = 49.89\ kJ/kg$$